

Boundary-railing of conventional NLLS fits as an assumption-free pseudo-diffusion identifiability diagnostic in IVIM MRI

A bound-pinned D^* failure mode, re-centred from a calibration ruler and replicated across open human-abdominal diffusion MRI

Abstract

Purpose. Reported uncertainty in intravoxel incoherent motion (IVIM) imaging has been critiqued as *overextended*: any coverage or calibration statement is only as trustworthy as the noise and forward model that generated its reference truth. We answer that critique not by defending the calibration argument but by promoting, as the primary contribution, a result that *cannot* be overextended because it assumes nothing. That D^* is weakly identified is itself long known; what is new is not the phenomenon but an assumption-free, optimiser-level statistic that neutralises this objection where no calibration ruler can, together with its first systematic, pre-registered, cross-cohort quantification and an independent clean-room reproduction.

Methods. In a conventional bounded non-linear least-squares (NLLS) bi-exponential IVIM fit, the pseudo-diffusion coefficient D^* is so weakly identified that the optimiser frequently terminates *on* a parameter bound. We read this boundary-railing directly off the optimiser’s output — no ground truth, no noise model — and quantify it with voxel-level bootstrap confidence intervals on one open human-abdominal acquisition (OSIPI), its full ROI, and an independent liver-DWI cohort (TCGA-LIHC). The calibration ruler (coverage, expected calibration error, sharpness) is retained but demoted to a scoped secondary diagnostic, evaluated only where ground truth exists, and under an explicitly stated honest Cramér–Rao (CRLB) convention.

Results. On the curated homogeneous ROI, 54.7% of 1,618 high-SNR voxels rail D^* (95% CI [52.2, 57.1]), reproducing a prior report and corroborated by an independent clean-room reimplemention (54.2% [52.0, 56.4]). Railing generalises to the full abdomen (47.8%), survives generous bounds and every SNR stratum, and replicates on liver DWI (43.7% at a clean 4-*b* scheme; 73.4% across 3 subjects at a sparse 3-*b* scheme). As a scoped secondary finding, under the honest CRLB symmetric Gaussian intervals under-cover D^* specifically in the high- D^* tercile (central-95% coverage 0.63 [0.59, 0.67] for the Laplace SD), and an amortized neural posterior is simultaneously better-calibrated and sharper than the railed-NLLS baseline.

Conclusion. A large, reproducible fraction of conventional NLLS IVIM fits rail to a parameter bound on open human-abdominal data. Because the statistic is read off the optimiser without any calibration-trust assumption, it is immune to the overextension objection; the calibration ruler remains useful but scoped.

I Introduction

The clinically load-bearing question in quantitative diffusion MRI is not how accurate a point estimate is, but whether a reported *error bar* can be believed. The intravoxel incoherent motion (IVIM) model [1] separates tissue diffusion from microcapillary perfusion through a bi-exponential signal decay, and its perfusion-sensitive pseudo-diffusion coefficient D^* is both the clinically interesting parameter and the notoriously ill-conditioned one [2, 3]. A natural way to certify an uncertainty estimate is a calibration ruler — empirical coverage, expected calibration

error (ECE), and sharpness at fixed nominal levels — that scores whether a method’s stated uncertainty is honest. A growing literature applies exactly such rulers to classical Bayesian and learned IVIM estimators [4, 5, 6, 7, 8].

The objection a sceptical reviewer raises against any such ruler is that it is itself *overextended*. Empirical coverage is computed against a reference truth simulated under an assumed noise and forward model, so a coverage claim inherits exactly the trust it was meant to certify: if the simulator is misspecified, the calibration verdict is too. This is not a hypothetical concern — model misspecification is the central failure mode documented for simulation-based inference [9, 10, 11], and a methods reviewer of an earlier version of this work flagged the calibration contribution as resting on precisely this unverifiable assumption.

This paper takes the objection seriously and re-aims the contribution. We promote to the primary claim a phenomenon that requires no calibration trust at all: **boundary-railing** of the conventional NLLS IVIM fit. When the likelihood is flat in D^* , a box-constrained optimiser does not find an interior optimum — it slides to a constraint and stops. The fraction of voxels for which this happens is a property of the estimator applied to the data; it is read directly off the optimiser’s output and carries no assumption about ground truth or noise. It therefore cannot be overextended. We are explicit about what is new. That D^* is weakly identified — and that a bounded fit can consequently pin it to a bound — is long established [2, 3]; the contribution is not that observation but an assumption-free, optimiser-level statistic that answers the overextension objection no calibration ruler can, given here its first systematic, pre-registered, cross-cohort quantification and an independent clean-room reimplementa-tion. We show that this railing fraction is large, reproducible, generalises beyond a hand-curated ROI, survives deliberately generous bounds and every SNR stratum, and replicates on an independent organ, scanner, and site.

The calibration ruler is not discarded. It is kept as a clearly scoped *secondary* diagnostic, evaluated only on data with ground truth, and reported *conditionally* on the true D^* regime under a stated honest CRLB convention — a deliberately narrower claim than the marginal headline an earlier draft carried. Within that scope it yields a real, reproducible finding: symmetric Gaussian intervals under-cover D^* in the high- D^* regime, the same weak-identifiability wall that drives railing. The contribution is thus a single mechanism — weak D^* identifiability — observed two ways: as an assumption-free fact about the optimiser (primary), and as a scoped calibration consequence where ground truth permits it (secondary).

2 Methods

This section is complete from line one: it names every dataset with a stable identifier, states every fitting and training detail, and documents both the Cramér–Rao approximation and the railed-voxel standard-deviation (SD) convention that earlier reviewers flagged as missing. A completeness checklist is given in Table 3.

2.1 Datasets

Primary real cohort — OSIPi abdomen. The primary in-vivo cohort is the open human-abdominal IVIM acquisition distributed with the OSIPi TF2.4 IVIM code collection (Zenodo record 14605039, DOI 10.5281/zenodo.14605039, CC-BY-4.0; Philips 3T) [12]. The archive provides a $144 \times 144 \times 21$ volume with 104 diffusion measurements over 12 unique b -values $\{0, 10, 20, 30, 40, 50, 75, 100, 150, 250, 400, 600\}$ s/mm². We analyse the archive’s homogeneous-tissue mask (1932 voxels; each voxel b_0 -normalised by its mean $b=0$ signal) and, separately, the full abdominal ROI. The homogeneous ROI is the cohort on which a previously reported 54.7% NLLS D^* -railing fraction was first computed.

Independent replication cohort — TCGA-LIHC liver. The independent cohort is liver diffusion-weighted imaging from the TCGA-LIHC collection (TCIA; DOI 10.7937/K9/TCIA.2016.IMMQW8UQ, CC BY 3.0; Siemens 1.5T) [13] — an unrelated site, scanner manufacturer, and organ. We use 4 series: a clean 4- b scheme ($b=0, 50, 500, 800$ s/mm²; one subject) and a sparse clinical 3- b scheme ($b=50, 400, 800$ s/mm², no $b=0$; 3 subjects).

Synthetic substrate for the ruler. The calibration ruler requires ground truth and is therefore evaluated only on a synthetic digital reference object. Ground-truth (D, D^*, f) are drawn from documented physiological priors; Rician noise is added at $\text{SNR} \in \{10, 20, 40\}$ with 200 noise realisations per condition (a 9-cell design). The substrate is generated by an independent read-only sibling module and is fully reproducible from its master seed.

Data and IP posture. All imaging is open and download-on-demand (OSIPI, CC-BY-4.0; TCGA-LIHC, CC BY 3.0), checksum-verified, and never redistributed in-tree; only provenance manifests are committed. No proprietary or clinical data are used.

2.2 Forward model and estimators

The bi-exponential IVIM model is

$$S(b)/S_0 = (1 - f) e^{-bD} + f e^{-b(D+D^*)}, \quad (1)$$

fit per voxel over (S_0, D, f, D^*) in scaled parameters so every fit variable is $O(1)$, with an analytic Jacobian verified against finite differences.

NLLS and boundary-railing. The point estimator is a box-constrained four-parameter trust-region-reflective least-squares fit (`scipy.optimize.least_squares`, analytic Jacobian, `max_nfev` = 400). The box follows the stated prior range. A parameter is *railed* iff $|\hat{x} - \text{bound}|/(\text{upper} - \text{lower}) < \varepsilon$; we report the D^* -railing rate at $\varepsilon = 10^{-3}$ (primary) and 10^{-2} (sensitivity), and under a deliberately widened box (“generous bounds”). The railing rate needs no ground truth: it is a property of the estimator applied to the data.

Laplace and MCMC posteriors. For the synthetic ruler only, two classical posteriors are formed: a Laplace/asymptotic Gaussian at the MAP with the CRLB covariance $\sigma^2(J^\top J)^{-1}$ (symmetric SD interval), and a clean-room random-walk Metropolis sampler (per-voxel, vectorised; Gaussian likelihood, known σ , uniform-box prior; burn 1500, keep 2000, thin 2), from which we read both a symmetric SD interval and a 2.5/97.5 quantile interval.

Amortized neural posterior (NPE). A conditional autoregressive normalizing flow (5 affine layers, per-dimension MLP conditioners, standard-normal base) [14, 15] is trained by neural posterior estimation on 80 000 (θ, x) pairs ($\theta \sim \text{uniform-box}$, $\text{SNR} \sim U(10, 40)$; Adam, learning rate 10^{-3} , 40 epochs, batch 512), written from scratch with no external SBI library.

2.3 The CRLB assumption and the railed-voxel SD convention

Two distinct modelling choices live here; both are stated, and the second is decisive for the calibration headline, so it is documented in full.

(a) *Gaussian/CRLB approximation.* The Laplace covariance $\sigma^2(J^\top J)^{-1}$ is a Gaussian approximation to the posterior, weakest precisely where D^* is most skewed — low SNR and high D^* , exactly where the under-coverage sits. The MCMC results do not rely on it (the sampler targets the exact posterior under the box prior).

(b) *Railed-voxel SD convention.* A voxel whose D^* is railed carries no local Fisher information about D^* : the Jacobian column collapses and $J^\top J$ is singular. Two defensible conventions then exist and give *opposite* coverage. Under the **honest CRLB** (our default) the pseudo-inverse assigns such a voxel a wide D^* interval — the statistically correct admission that D^* is undetermined there. Under an **overconfident floor** a railed voxel is instead given a small floored D^* SD, so its interval is narrow and under-covers “by design.” The convention is decisive: flooring drops pooled Laplace D^* coverage from 0.80 to 0.68, reconstructing a dramatic marginal figure that is an artefact of the convention rather than a property of the data. We report all calibration results under the honest CRLB and show the floored convention only as a labelled illustration (Section 3.4).

Cohort	n (high-SNR)	Railed	95% CI	Verdict
Abdomen, homogeneous ROI (OSIPI)	1,618	54.7%	[52.2, 57.1]	REPLICATES-STRONG
Abdomen, full ROI (OSIPI)	19,652	47.8%	[47.1, 48.5]	REPLICATES-STRONG
Liver, 4- b 0/50/500/800 (TCGA-LIHC)	648,265	43.7%	[43.2, 44.2]	REPLICATES-STRONG
Liver, 3- b 50/400/800 (TCGA-LIHC)	1,544,999	73.4%	[73.0, 73.9]	REPLICATES

Table 1: Boundary-railing of the NLLS D^* across open human-abdominal cohorts (seed 20260613, 5,000-resample bootstrap). Pre-registered thresholds: railed $\geq 30.0\%$ with bootstrap lower bound $\geq 20.0\%$. The homogeneous-ROI rate matches the prior report of 54.7% and is independently corroborated by a clean-room reimplementation (54.2% [52.0, 56.4]).

2.4 Calibration ruler and uncertainty on the numbers

The ruler — coverage, ECE (mean $|\text{empirical} - \text{nominal}|$ over a level grid), sharpness (mean interval width) — is re-derived from published definitions, with a D^* -tercile conditional-coverage probe added. Every load-bearing number carries a bootstrap confidence interval (voxel-level resampling for the real cohorts, 5,000 resamples, seed 20260613; noise-realisation resampling for the synthetic ruler). A directional gap is reported as significant only if its bootstrap CI excludes zero. Pre-registered replication thresholds (fixed before running): a cohort *replicates* if the railed fraction is $\geq 30.0\%$ *and* its 95% bootstrap lower bound is $\geq 20.0\%$.

3 Results

3.1 Primary: boundary-railing across cohorts (assumption-free)

On the curated homogeneous ROI, 54.7% of the 1,618 high-SNR voxels have a bound-railed NLLS D^* (95% CI [52.2, 57.1]), matching the previously reported 54.7% on the same data (verdict REPLICATES-STRONG). An independent clean-room reimplementation — a separate forward model, fitter, and railing test sharing no code with the original — obtains 54.2% ([52.0, 56.4]) on the same ROI, and 58.7% on a stricter b_0 -SNR >25 subset; the two CIs overlap, so the phenomenon is not an implementation artefact.

Railing is not a property of a hand-chosen homogeneous patch. Extending to the entire abdominal ROI ($n=19,652$), the rate is 47.8% ([47.1, 48.5], verdict REPLICATES-STRONG). It also recurs on an independent organ, scanner, and site: on the clean 4- b liver scheme the railed fraction is 43.7% ([43.2, 44.2], $n=648,265$, verdict REPLICATES-STRONG), and across 3 subjects at the sparse 3- b clinical scheme it is 73.4% ([73.0, 73.9], $n=1,544,999$, verdict REPLICATES). All four cohorts clear the pre-registered replication thresholds (Table 1).

3.2 Mechanism: why D^* rails

Railing is structural weak identifiability of D^* , not measurement noise. It does not vanish with SNR — it persists across every SNR stratum — and it survives deliberately generous bounds, falling but remaining a substantial minority (32.1% on the homogeneous ROI, 27.4% on the full abdomen), so it is not an artefact of tight box constraints. It is dominated by the *upper* D^* bound: the homogeneous ROI splits 15.6% lower-rail / 39.1% upper-rail, and the full ROI 4.6% / 43.2%. This is the high- D^* identifiability wall: the fast perfusion compartment is so weakly constrained by the data that the optimiser is pushed to the ceiling. The direction of the effect across acquisition schemes confirms the mechanism — the sparser the low- b perfusion sampling, the worse D^* identifiability, and the more the fit rails (the sparse 3- b liver scheme rails most). Equation (1) makes this concrete: D^* governs a fast decay visible only at low b , so once the low- b sampling is thin the likelihood is nearly flat in D^* and the asymptotic (CRLB) variance — which is exactly the curvature the optimiser cannot find — diverges.

Estimator (honest CRLB)	low D^*	mid D^*	high D^*	pooled
Laplace SD	0.91 [0.89, 0.94]	0.86 [0.83, 0.89]	0.63 [0.59, 0.67]	0.80 [0.78, 0.82]
MCMC SD	0.95 [0.93, 0.97]	0.95 [0.94, 0.97]	0.81 [0.78, 0.84]	0.90 [0.89, 0.92]
MCMC quantile	0.93 [0.90, 0.94]	0.97 [0.95, 0.98]	0.81 [0.78, 0.84]	0.90 [0.89, 0.91]

Table 2: Per-true- D^* -tercile central-95% D^* coverage on the synthetic substrate, under the honest CRLB convention (Section 2.3), with bootstrap CIs. The under-coverage is conditional — concentrated in the high- D^* tercile — and survives the quantile-shape fix.

3.3 Secondary, scoped: the calibration ruler under the honest CRLB

The same weak-identifiability wall has a calibration consequence, which we report only where ground truth exists (the synthetic substrate) and only conditionally on the true D^* regime. This is the deliberately scoped secondary contribution; it is undefined on the real abdominal scan, which has no ground-truth (D, D^*, f), and that limitation — the very point the overextension critique raised — is the argument for leading with railing.

Symmetric Gaussian intervals under-cover D^* in the high- D^* regime. Under the honest CRLB, central-95% D^* coverage is near-nominal in the low- D^* tercile but falls in the high- D^* tercile: 0.63 [0.59, 0.67] for the Laplace SD and 0.81 [0.78, 0.84] for the MCMC SD, versus 0.91 and 0.95 respectively in the low- D^* tercile (Table 2). The failure is identifiability-driven and conditional — it is concentrated exactly where railing concentrates — not a uniform marginal collapse.

The interval shape, not a wider SD, restores marginal coverage. Reading the MCMC posterior’s 2.5/97.5 quantile interval rather than a symmetric SD interval restores near-nominal *marginal* D^* coverage (0.90 [0.89, 0.91]); D and f are already near-nominal (0.95, 0.96). The right interval *shape*, not a larger SD, is the fix. Conditionally, however, a residual high- D^* gap remains even for the quantile interval (0.81 [0.78, 0.84] in the high- D^* tercile): the residual is the identifiability wall itself, not an interval-shape artefact.

An amortized posterior beats the railed NLLS on calibration and sharpness. On the synthetic held-out set the neural posterior is simultaneously better-calibrated and sharper than the railed-NLLS baseline: D^* coverage 0.98 versus 0.76 (gap 0.22 [0.20, 0.23]), ECE 0.069 versus 0.121, and sharpness 0.112 versus 0.181. All three gaps’ bootstrap CIs exclude zero. The railed Gaussian’s wide-but-miscentred intervals lose on every axis: railing is not merely a point-estimate pathology but an uncertainty pathology that a calibrated estimator avoids.

3.4 The convention reconstructs — but does not justify — a dramatic marginal figure

For completeness we document how a much more dramatic marginal under-coverage can be produced. Under the overconfident floored convention (Section 2.3b), pooled Laplace D^* coverage drops to 0.68 [0.65, 0.70] and the high- D^* tercile to 0.41 [0.37, 0.45]. Flooring an unidentified voxel’s SD reports D^* as if it were precisely known — it manufactures severity. We therefore do *not* report a marginal headline coverage; the honest, reproducible statement is the conditional table above, and the floored figures appear only here, with the convention named, to explain how a larger number can arise.

4 Discussion

What is and is not claimed. The primary claim is narrow and assumption-free: a large, reproducible fraction of conventional NLLS IVIM fits rail D^* to a bound on open human-abdominal data, across ROI, bounds, SNR, and an independent liver cohort. The secondary claim is scoped to ground-truth data: under a stated honest CRLB,

symmetric Gaussian intervals under-cover D^* in the high- D^* regime, a quantile interval fixes the marginal but not the conditional gap, and an amortized posterior dominates the railed NLLS. We explicitly do *not* claim a dramatic marginal coverage collapse: such figures are reconstructible only under an undocumented floored SD convention and are reported here solely as a labelled illustration. We also make no claim about an out-of-distribution self-consistency gate, estimator timing, or in-vivo brain held-out- b recovery; those questions are outside the scope of the evidence presented and are left to future work.

Differentiation from known D^* instability. That D^* is unstable and hard to fit is textbook [2, 3]; the contribution here is not that observation but an assumption-free, optimiser-level railing statistic — immune to the overextension critique that calibration claims invite — delivered with its first systematic, pre-registered, cross-cohort quantification and an independent clean-room reproduction.

Differentiation from learned uncertainty. Casali et al. [8] characterise one supervised network’s aleatoric/epistemic uncertainty budget, validated on preclinical mouse brain; this work instead documents an estimator-agnostic, real-data identifiability signature in human-abdominal DWI, their own residual D^* overconfidence being the learned echo of the same wall that rails the conventional fit.

Limitations and scope. The open primary cohort is a single subject; we make no population claim from it, reporting a within-subject voxel statistic with a voxel-bootstrap CI and supporting generality by cross-cohort agreement (homogeneous ROI, full ROI, and an independent liver cohort). The 3- b liver scheme lacks $b=0$ and is normalised by its lowest b , an approximation flagged as such; the 4- b liver cohort with $b=0$ is the clean independent test. Railing magnitude depends on the bound width and the b -scheme density; we report both tight and generous bounds and per-scheme results rather than a single number. The calibration ruler is evaluated only on synthetic ground truth and is, by construction, undefined on the real scans — which is exactly why it is the secondary, not the primary, contribution.

5 Conclusion

A large, reproducible fraction of conventional NLLS IVIM fits rail to a parameter bound on open human-abdominal data, and the effect generalises across ROI, bounds, SNR, and an independent liver cohort. Because it is read off the optimiser without any calibration-trust assumption, boundary-railing is a primary claim immune to the overextension objection that calibration arguments invite. The calibration ruler remains useful but scoped: under an honest CRLB convention it shows a real, conditional high- D^* under-coverage and an amortized-posterior advantage, with every number traced to a seeded, openly reproducible computation.

Data availability and reproducibility

All imaging is open and download-on-demand: the OSIP abdomen acquisition (Zenodo 14605039, CC-BY-4.0) and the TCGA-LIHC liver series (TCIA, 10.7937/K9/TCIA.2016.IMMQW8UQ, CC BY 3.0), each checksum-verified and never redistributed in-tree. Every number in this manuscript is regenerated from seeded result files by a single consistency gate that fails the build on any drift; the synthetic substrate, the railing analysis, the classical and amortized posteriors, and the calibration ruler are each independent, seeded reimplementations.

References

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Completeness item	Where addressed
Dataset identifiers	§2: OSIPi Zenodo 14605039; TCGA-LIHC 10.7937/K9/TCIA.2016.IMMQW8UQ
Acquisition / b -schemes	§2: 12 unique b ; liver 4- b and 3- b schemes
Full fitting (NLLS, MCMC)	§2: box/init/solver/railing; sampler spec
Full training (NPE)	§2: architecture, 80k sims, prior, epochs
CRLB approximation	§2.3(a)
Railed-voxel SD convention	§2.3(b); illustration §3.4
Calibration ruler definitions	§2
Uncertainty / CIs	§2: bootstrap, seed 20260613
Claims scope (no overclaim)	§3.1–3.4; Discussion

Table 3: Completeness checklist — the methods items an earlier review flagged as missing, each mapped to where it is now addressed.

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